How the spaceplane autopilot works

krpcLab · spaceplane · what the program does and what makes it change what it is doing. State of the code on 2026-10-21 (defaults fingerprint `cb0ae1d4`).

The autopilot takes a winged vehicle (the target is the Mk3 shuttle `qs_shuttle2`) from orbit to a stop on the KSC runway. It does not follow a stored plan. Every tick it reads the vehicle's state, propagates the rest of the flight to the ground with an RK4 integrator (`trajectory.py`), and takes each decision from that fresh prediction. `autopilot.py` is the phase machine and the only code that talks to the game; `guidance.py` holds the control laws as pure functions; `environment.py` holds the aerodynamic table the vehicle measures for itself; every tunable is in `config.py`.

The phase machine

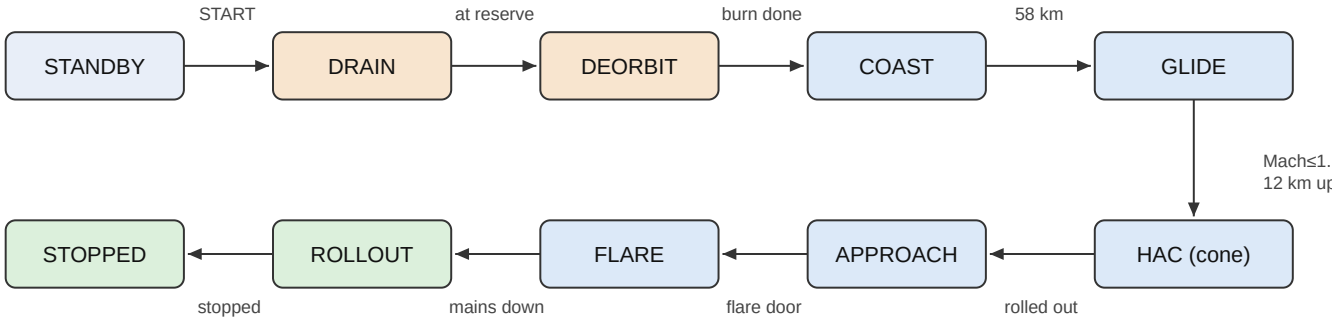


Figure 1. Phases and the conditions that advance them. Orange phases use the engine; blue are unpowered flight; green is on the ground. Engaged below the interface, START skips to GLIDE, APPROACH or FLARE by height and distance to the runway.

Things that run in every phase

mechanism	what it does, and when
Prediction	A propagation to the ground of the law the vehicle will fly (the same function the controller calls), from the current state, each tick. The deorbit and the glide solve against it; the cone and the approach plan from it.
Aerodynamic table	Swept in STANDBY with kRPC's `simulate_aerodynamic_force_at` over angle of attack −8° to 90° and Mach; rows are re-probed in flight. Lift is **signed** on the rows below zero alpha (the component along the normal of the pitch plane), and a table with such rows is read signed rather than mirrored, so a cambered wing's polar is measured on both sides of zero. Stall speed (an equivalent airspeed), best L/D and the approach speed are read from it, not configured.
Attitude	kRPC's autopilot points the nose at the commanded (alpha, bank). Its response times are derived from the vessel's inertia and torque and retuned as surface authority grows with dynamic pressure. A roll damper lengthens the roll response when bank swings grow and recovers it after.
Bank slew	In GLIDE, HAC and APPROACH the bank command never steps (`guidance.ease_bank`): it moves no faster than the measured roll rate, accelerates and brakes at half the bank acceleration the wheels' and control surfaces' torque allows at the alpha flown (RCS torque is left out, so the slew does not depend on whether the valve happens to be open), and arrives at the new lean with no rate to carry past it.
RCS valve	Burn and coast (including the coast's turn from the burn attitude to the entry attitude after the warp): open when the pointing error exceeds 5°, closed 2 s after it is under 1.5°. GLIDE through FLARE: armed, and fires on saturation — once an axis's summed control input has sat at ≥ 0.95 for 0.5 s while the pointing error has not closed by 20%; closed once the input has stayed under 0.6 for 2 s. A bank reversal holds it open. Shut above 20 kPa.
Alpha ceiling	Learned: when the vehicle sits well below its commanded alpha, the ceiling ratchets down to what it holds, and the prediction uses the same limit (`trajectory.Holdable`, binned by dynamic pressure).

Pitch trim	Above Mach 3 in the glide the standing pitch input is moved into a body-frame manual trim (`GLIDE_PITCH_OFFLOAD`). In the cone and approach the nose canards are driven as a trim that walks the standing pitch input to zero (`CANARD_TRIM`).
Brakes	Built from the control surfaces and probed in STANDBY: the split rudder (the two fins opposed: drag with no yaw or roll) and the spoiler (opposed flaps). Used only below the glide, in the order split rudder → spoiler → weave/S-turn.
Time scale	The farm's time-scale plugin is set so each tick's game interval stays what the phase asked for; every log ends with a `loop rate` line.

Phase by phase

STANDBY — until START is pressed

Configures the craft (main gear brakes 200% / friction 10, nose gear brake off), measures wheel clearance and tail-strike angle, tunes the attitude controller, sweeps the aerodynamic table, and finds the brakes it can build from the surfaces. On START: below the 58 km interface → GLIDE / APPROACH / FLARE by height and distance; otherwise → DRAIN.

DRAIN — before the burn

Opens the drain valves and empties propellant down to a reserve of 200 m/s of burn (or 1.25 × the burn that puts the periapsis at the surface, if more), at the engine's vacuum Isp. → DEORBIT.

DEORBIT — the latest window, line up, one burn

- **Search** (`deorbit_solution`): a grid then refinement over retrograde dv 10–400 m/s, as an impulse. For each candidate it propagates the two corners of what the glide may later command (`deorbit_window`): the shortest entry (high drag) and the longest (most lift). The burn chosen puts the gate 25% of the half-width short of the window's centre. Each corner's peak heating index ($\sqrt{\rho}\cdot v^3$) and the window's width are kept with the answer.
- **Plan, first window.** On entry the orbit is propagated forward in vacuum and the search is asked of future states every 60 s (bisected to 8 s), up to three of Kerbin's days ahead, 1.5 s of work a tick. A state qualifies when the runway is within the glide's reach along track (the arc from the interface to the gate at most 1400 km) and sideways (the gate's offset from the ground track at most half the reach of an entry leaning one way), and its burn costs no more than the burn at apoapsis that puts the periapsis at the surface (a dearer one is remembered and the scan skips half an orbit; if the horizon runs out the cheapest seen is taken).
- **Plan, then the latest** (`DEORBIT_LATEST`, Figure 2). From the first qualifying state the scan keeps stepping forward while a burn still qualifies *and* (a) its window is at least 15% as wide as the first window's — the glide keeps room to correct — and (b) its hotter corner's peak heating is no more than the larger of 650 and the first window's own, and (c) its dv is no more than a surface periapsis costs (the same bound the first-window scan holds). A failing step is bisected to 8 s. The plan is the latest qualifying state at least 60 s inside the last one. While the walk runs it owns the tick: the live search does not take the first hit meanwhile, and an attitude hold begun before the plan is released so the lead re-takes it on the plan's node. Later burns are closer to the runway and larger: on the 80 km orbit about 62 m/s, burn to stop about 1100 s.
- **Wait.** A stock maneuver node is placed at the window; `warp_to` takes the game to 60 s before it. In that minute the live search runs every tick but does not commit before the planned time. A window not seen 120 s after it was due is replanned.
- **Burn lead.** The live trigger asks the search of the state half a burn ahead and commits when that solves, so the finite burn straddles the impulsive point. Duration $\max(3\text{ s}, dv\cdot m / F_{vac})$.
- **Line up.** The game's SAS holds Maneuver, RCS on; the burn commits once the nose is within 3° on two ticks in a row, or after a hold stuck inside 8° for 6 s, or after 120 s.
- **Burn.** SAS holds the committed attitude; thrust is limited so the burn lasts at least 3 s, tapered on the dv still owed; the stop test re-propagates the arc each tick. A runaway guard ends the burn 60 s after ignition whatever the attitude. At cutoff → COAST.

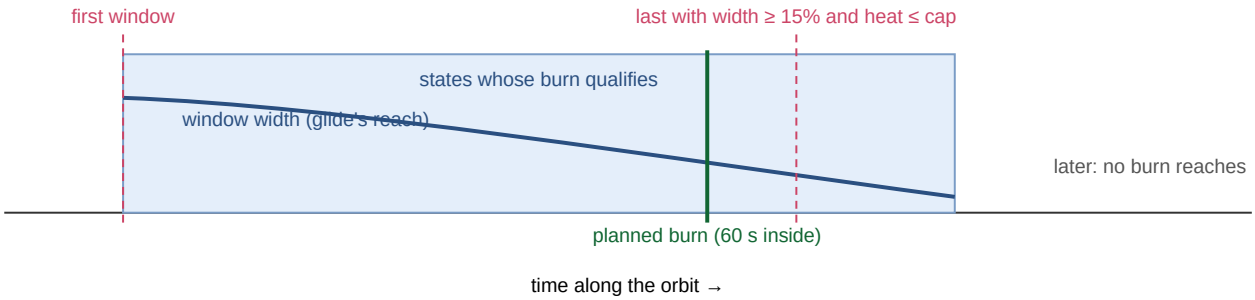


Figure 2. The latest-window rule. Every state between the first window and the end of the band could reach the runway; later ones are closer and need a larger, steeper burn, which leaves the glide less room. The plan takes the latest state that still leaves 15% of the first window's width and does not run hotter than the cap.

COAST — to the 58 km interface

`warp_to`s the part above the atmosphere (the vehicle keeps the burn's retrograde attitude through the warp), then swings to the entry attitude (22° alpha, 30° bank toward the runway) in the last minute above the air, with the RCS valve permitted (`COAST_RCS_TURN`): a turn of up to ~155° in under a minute is beyond the reaction wheels. The bank's side is latched so the glide starts on the same lean. Its first tick pumps the propellant left after the burn into each kind's forward-most tank (`FUEL_TO_NOSE`). At 58 km → GLIDE.

GLIDE — the entry, Mach 25 to Mach 1.5

Each tick (1 s): solve alpha and bank against the prediction. Above Mach 4.5 the solve nulls the along-track miss where the arc crosses 12 km; at Mach ≤ 4.5 (latched) it nulls the energy handed to the cone — the predicted energy height h + v²/2g at the cone's entry against 12 km at the gate speed plus a 1500 m margin (≈ 14.1 km).

control	role
alpha (20–40°)	Downrange / energy: more alpha, more drag. Rate-limited to 3°/s, held still during a reversal, clamped by speed and by the learned ceiling. In the terminal glide the prediction flies a 10 s mean of the achieved alpha.
bank magnitude (30–70°)	Takes what alpha leaves: rolling lift off vertical sinks the vehicle into denser air.
bank sign	Cross-track: lean toward the runway; reverse when the track's angle to the gate or the predicted cross-track leaves its deadband.

At Mach ≤ 1.5 and either 12 km over the field or within 3 km of the gate: release the alpha ceiling, pick the runway end and turn direction (`hac_choose`) → HAC.

HAC — the heading-alignment cone

- **Radius.** The circle (tangent to the extended centreline, ending at the 2 km gate) whose path, priced down a ladder of 500 m slices at the L/D the vehicle flies there, spends the energy above the gate (height plus the speed above the gate's target). 2–16 km, rate-limited; a lap only when even the widest circle is short of path.
- **Speed** (Figure 3). The target is the larger of two speeds. (1) Constant dynamic pressure (`HAC_SPEED_Q`): the speed whose q equals the gate target's (the approach speed, 2.25 × the measured stall) at the height flown. (2) The energy speed (`HAC_ENERGY_SPEED`, `guidance.cone_energy_speed`): the fastest speed whose own L/D (the table, times the measured L/D scale) spends 90% of the energy above the gate over the tightest circle the vehicle can hold at that speed (45° of bank, 1.3 × the level-turn radius) — faster lands short, slower leaves surplus for the brakes. It is capped where lift at the −5° alpha floor still carries the weight (the airframe's own top speed; no Mach limit) and by what drag at full alpha and split rudder could still bleed to the gate's speed over the path left. Not used when the cone is short of energy.
- **Alpha holds the speed:** the trim alpha for the banked load plus 0.2° per m/s of excess (clamped to ±25 m/s), −5° to 22°, capped at 3 g of lift.
- **Split rudder, first:** opens to the angle whose L/D factor spends the height to the gate over the planned path; fades only near the stall. **Spoiler, second; weave, last** (up to 50° off track when more than 800 m of surplus remains and the brakes cannot spend it).

- **Exit.** Lined up, within 900 m of the gate, no lap left, no more height than the approach can absorb → APPROACH. Below 2 km → APPROACH regardless.

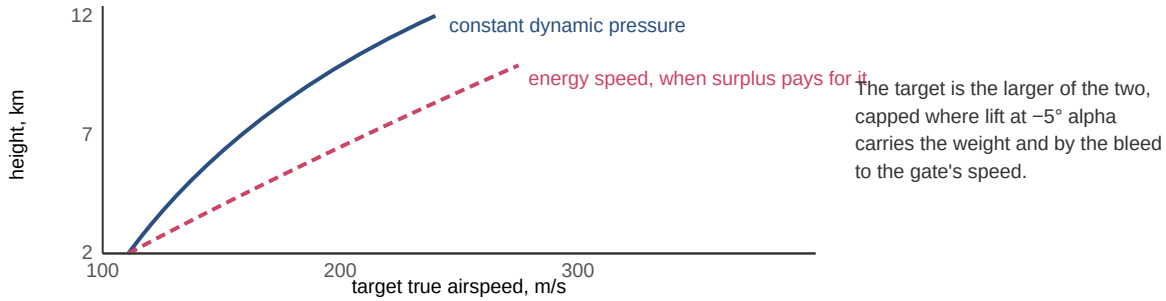


Figure 3. The cone's speed target against height (shape only). On the shuttle the q target is about 112 m/s true at the 2 km gate and 245 m/s at 12 km; the energy solve asks for roughly the same speed, because the cone arrives with little surplus.

APPROACH — final, 2 km to the flare

- **Aim point:** 1200 m past the threshold (`TOUCHDOWN_AIM_M`); the approach aims 1000 m nearer so the flare's float lands in the zone — touchdown about 600–800 m past the threshold.
- **Speed, not path:** alpha is set from the descent angle that holds the target speed (2.25 × stall), from dv/dt = g sinθ − D/m. The trim comes from the table (now including negative alpha); the command is floored at 0°.
- **Split rudder:** opens to the L/D factor the geometry asks for; **spoiler** when it is at its stop; **S-turns** when more than 200 m of excess height remains and both brakes are at their stops.
- **Lateral:** a closure that brings cross-track to zero as the wheels arrive, bank floor 6°.
- **Divert north** (`ABORT_NORTH`): if the energy height (height plus the speed above best-glide speed, as $(v^2-v_{bg}^2)/2g$) over the distance to the threshold at the swept table's best glide ratio (at the landing mass, never below the approach's 4.2) falls short by more than 10% of the height left (never under 30 m), the tracked line is latched 150 m north (the space center lies south).
- **Flare door:** 50 m + 2.5 s × sink rate → FLARE.

FLARE — the last 50–100 m

Stows the brakes. Tracks a sink rate that eases to the touchdown rate, with alpha capped by maximum lift and by the tail-strike angle; below 30 m the nose is aligned with the runway. Gear down. Main wheels touching → ROLLOUT.

ROLLOUT — on the runway

Ground spoiler out on contact; the nose ramped down over 1.5 s; full main-gear braking; nosewheel steering onto the centreline. If the stopping distance — at the larger of the last second's deceleration and the average since touchdown — runs past the far end, the line steered to is latched 150 m north. At the stop speed → STOPPED, which logs the stop and hands the vessel back.

Which law decides what

quantity	DEORBIT	GLIDE	HAC	APPROACH	FLARE / ROLLOUT
when / where	latest window with ≥ 15% of the first's reach, heat ≤ cap, dv ≤ a surface periapsis	—	—	aim 1200 m past the threshold	—
downrange / energy	burn dv	alpha, then bank: range above Mach 4.5, cone-entry energy below	radius and laps; split rudder → spoiler → weave	split rudder → spoiler → S-turns; divert north when short	divert north on an overrun
speed	—	speed cap on alpha	alpha to max(q target, energy speed), −5° floor	descent angle that holds speed	sink-rate law / wheel brakes
cross-track	sideways reach test	bank sign deadbands	circle geometry	closure, bank floor 6°	bank ≤ 12°, nosewheel

Logs

Nothing prints. Each run writes `logs/LOG<n>`: a `config:` line (fields different from the defaults, plus the defaults fingerprint), one telemetry line per tick interval, every phase change and decision as an event, and a closing `loop rate` line. The deorbit logs `deorbit latest:` (the span of qualifying burns, the plan, the dv at the first and the planned window, the heating) and the window's width at commit. The cone's line carries `vt=` (the speed alpha flies to) and `ves=` (the energy speed). The `DOWN:` line gives the stop along and across the runway and whether it is on the tarmac — the measure of a landing.